

DR. RYSSA MOFFAT

CONTACT

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Social Brain Sciences Lab
ETH Zurich

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ACADEMIC APPOINTMENTS

2023–present **Postdoctoral Researcher**
ETH Zurich, Department of Humanities, Social and Political Sciences
Social Brain Sciences Lab, Director: Prof. Emily S. Cross

2021–2023 **Postdoctoral Research Fellow**
Macquarie University, Faculty of Medicine, Health and Human Sciences
Social Brain in Action Lab, Director: Prof. Emily S. Cross

EDUCATION

2022 **Macquarie University, Newcastle University, Universities of Groningen & Potsdam**
PhD in Cognitive Science (joint degree)
Thesis: *Recognition and cortical haemodynamics of vocal emotions – an fNIRS perspective*
Committee: Prof. David McAlpine, Prof. Deniz Başkent, Prof. Lindsey van Yper, Dr. Robert Luke

2018 **Universities of Eastern Finland, Groningen & Potsdam**
M.Sc. in Clinical Linguistics (joint degree)
Thesis: *Coarticulation as a synchronic predictor of reading dysfluency*
Committee: Prof. Martijn Wieling, Prof. Aude Noiray

2016 **University of Ottawa**
HBA, Major in German Language and Culture, Minor in Linguistics
Exchange year at University of Bonn

RESEARCH VISION

To foster social wellbeing for people of all ages, **I synthesize multi-modal data, merging mobile neuroimaging of single or multiple brains, motion tracking, and hormone tracking with behavioral and experiential perspectives**. By capturing and integrating these diverse streams, I aim to uncover the underlying mechanisms of thriving social connections and develop evidence-based frameworks to bolster social wellbeing across the lifespan.

HONOURS

2026 **fNIRS Open Dataset Contest (2000 USD)**
Awarded 1st place for InterGenSynchrony Dataset.

2026 **Sex as a Biological Variable COST – Short Term Scientific Mission Grant (1000 EUR)**
Awarded for research visit to Prof. Falkner's neuroendocrinology lab at Princeton University.

2025 **Swiss Academy of Humanities and Social Sciences (1000 CHF)**
Awarded for international conference travel.

2022 **Australasian Society for Social and Affective Neuroscience (100 AUD)**
Awarded for national conference travel.

2019 **Conference on Implantable Auditory Prostheses – Travel Award (1000 USD)**
Awarded for international conference travel.

2016 **EMCL Consortium – EMCL+ Scholarship for M. Sc. in Clinical Linguistics (2000 EUR)**
Awarded to cover international tuition fees.

GRANT SUPPORT

2026	**ETH Zurich Career Seed Grant – Pending, proposal under evaluation (100,000 CHF)**
2026	**SNSF Ambizione – Finalist, invited to interview (750,000 CHF)** Scores above 9/10 from all four external reviewers.
2023	**Cortivision Pathfinder Program – Neuroimaging equipment loan (valued at 86,000 EUR)** Awarded for project titled "Probing the development of behavioural and neural synchrony in intergenerational interactions via collaborative art tasks". Award consisted of loan of two fNIRS devices for 9 months.
2023	**Caran D'Ache – Art materials donation (valued at 1000 EUR)** Award of 30 sets of NEOCOLOR pastels to be used in experimental research involving artmaking.
2019	**Macquarie University – Postgraduate Research Funding (6000 AUD)** Awarded for studies using fNIRS to record brain activity evoked by vocal emotions.
2018	**IDEALAB Consortium – IDEALAB PhD Fellowship (160,000 EUR)** Awarded for a 3-year PhD project investigating the behavioural and neural correlates of emotion recognition in natural and degraded speech. Covered experimental work, publication fees, relocation between countries, 2 Australia-Europe travel allotments per year for training and research dissemination.

PUBLICATIONS & PRESENTATIONS

* indicates equal contribution ^ indicates mentored student

PREPRINTS

1. **Moffat** & Cross. (2026). A longitudinal fNIRS hyperscanning dataset of same generation and intergenerational relationship development and collaboration. *bioRxiv*.
2. Casale^, **Moffat**, Demir^, Nagel^ & Cross. (2026). Designing an intergenerational Dance for Health programme: Perspectives from younger and older urban Swiss adults. *PsyArXiv.*
3. Leisten^, **Moffat**, Caruana, & Cross. (2026). "If we are good friends, AI doesn't spy so much": Children's knowledge and misconceptions of AI safety. *PsyArXiv.*
4. Leisten*^, **Moffat***, Caruana & Cross. (2026). Children's understanding of sustainability and social equity of educational robots: Advice for educators and policymakers. *PsyArXiv*.

PEER-REVIEWED ARTICLES

1. **Moffat**, Dumas & Cross. (2026). Social interactions between people of same and different generations shape longitudinal changes in interpersonal neural synchrony, loneliness, and social connection. *PLOS Biology*.
2. Naudszus*^, **Moffat*** & Cross. (2026). Two-brain states characterise within- and between-brain connectivity during intergenerational collaborative drawing. *Acta Psychologica.*
3. Leipold & **Moffat**. (2026). Individual-specific precision neuroimaging of learning-related plasticity. *Neuroscience and Biobehavioral Reviews*.
4. **Moffat**, Naudszus^ & Cross. (2026). Cardiac synchrony during collaborative drawing: A longitudinal comparison of same generation and intergenerational dyads. *Annals of the New York Academy of Sciences.*
5. Pennington, Norris, Skubera, Ba, **Moffat**, Bohrer Amaral, et al. (2026). Awareness and use of open research practices: An international survey of researchers across disciplines. *Royal Society Open Science.*
6. **Moffat** & Cross. (2025). Individual abilities to estimate levels of movement synchrony predict action observation network activation. *Imaging Neuroscience.*
7. **Moffat***, Cahill*^, Cross & Caruana. (2025). Children with and without reading difficulty value robot reading companions that are smart, supportive, and personal. *Scientific Reports.*
8. Casale*^, **Moffat*** & Cross. (2024). Aesthetic evaluation of body movements shaped by embodiment and arts experience: Insights from behaviour and fNIRS. *Scientific Reports*.
9. Cross*, Darda*, **Moffat***, Muñoz, Humphries & Kirsch. (2024). Mutual gaze and movement synchrony boost observers' enjoyment and perception of togetherness when watching dance duets. *Scientific Reports*.
10. **Moffat** & Cross. (2024). Awareness of embodiment enhances enjoyment and engages sensorimotor cortices. *Human Brain Mapping*.

11. **Moffat**, Roos^, Casale^ & Cross. (2024). Dyadic body competence predicts movement synchrony during the mirror game. *Frontiers in Human Neuroscience*.
12. **Moffat**, Caruana & Cross. (2024). Inhibiting responses under the watch of a recently synchronized peer increases self-monitoring: Evidence from functional near-infrared spectroscopy. *Open Biology*.
13. **Moffat** & Cross. (2024). Evaluations of dyadic synchrony: Observers' traits influence quantification and enjoyment of synchrony in mirror-game movements. *Scientific Reports*.
14. **Moffat**, Casale^ & Cross. (2024). Mobile fNIRS for exploring inter-brain synchrony across generations and time. *Frontiers in Neuroergonomics*.
15. **Moffat**, Başkent, Luke, McAlpine & van Yper. (2023). Cortical haemodynamic responses predict individual ability to recognise vocal emotions with uninformative pitch cues but do not distinguish different emotions. *Human Brain Mapping*.
16. Caruana, **Moffat**, Blanco & Cross. (2023). Perceptions of intelligence & sentience shape children's interactions with robot reading companions. *Scientific Reports*.

CONFERENCE PROCEEDINGS

17. Leisten*^, **Moffat***, Caruana, & Cross. (2026). Children with greater sustainability and equity understanding demand more from educational robots. *Mindful TEL: Learning Technologies Shaped with Intention.*
18. Caruana, **Moffat**, Blanco & Cross. (2022). Talk, listen and keep me company: A mixed methods analysis of children's perspectives towards robot reading companions. *Proceedings of the 10th International Conference on Human-Agent Interaction*.

OTHER PROFESSIONAL PUBLICATIONS

19. **Moffat** & Cross. (2026). *InterGenSynchrony Dataset*. [Dataset]. OpenNeuro.
20. PPSP Team (contributor). (2026). Ppsp-team/HyPyP. [Python Package]. PPSP.
21. Preregistration Task Force (contributor). (2026). Preregistration Standards for Psychology - the Psychological Research Preregistration-Quantitative (aka PRP-QUANT) Template. *PsychArchives*.
22. Paquet, Prosser, **Moffat**, Davies, Degen & Sleigh. (2025). Open Science: A guide to recovery for activist burnout. [Zine]. *Research Equals*.
23. **Moffat**. (2024). Invisible mechanisms of interpersonal alignment. *Nature Reviews Psychology*.

INVITED TALKS

1. *Building intergenerational social connections through collaborative creative encounters*. Talk given to the International Centre for Community-based Loneliness Prevention, Tohoku University, Oct 2026.
2. *What does it take to initiate face-to-face interactions with unfamiliar people?* Talk given at "Social Connections with AI Agents Workshop", Collegium Helvetica, Oct 2026.
3. *Human brains and hearts: How people align during collaboration.* Talk given at Princeton Neuroscience Institute, Princeton University, Sep 2026.
4. *Quantifying connections: Movement and interbrain synchrony as metrics of social connection.* Talk given at Research Unit for Developmental Psychology, University of Vienna, Mar 2026.
5. *Evolving social connections: Three approaches to analysing longitudinal hyperscanning data*. Talk given at "In-Sync, a Hyperscanning Workshop", Inria, Feb 2026.
6. *From strangers to friends: Mapping the development of intergenerational relationships with multi-brain fNIRS.* Talk given at the Faculty of Psychology and Educational Sciences, University of Geneva, Oct 2025.
7. *The beauty of moving together – Using light to understand synchrony between bodies and brains.* Talk given at "Exploring the Spectrum of Light Workshop", Collegium Helvetica, Jun 2025.
8. *From speech processing to complex real-world social interactions using fNIRS.* Talk given to the Evolutionary Neuroscience of Language & Computational Neuroscience of Speech & Hearing research groups, University of Zurich, May 2025.
9. *Hyperscanning: Practicalities & Opportunities*. Talk given at "fNIRS Workshop – Hyperscanning & Developmental Studies", University of Vienna, Sep 2024.
10. *Interpersonal synchrony in movement and the brain*. Talk given to the Precision Psychiatry & Social Physiology Team, CHU Sainte-Justine Research Center, Apr 2024.

11. *Neuroaesthetics and Joint Action Perception*. Talk given to the Behaviour Section of the Department of Humanities, Social and Political Sciences, ETH Zurich. Dec 2023.
12. *Towards an fNIRS paradigm to map emotional prosody processing*. Talk given to the CI Brain Lab, Sunnybrook Health Science Centre, Feb 2020.
13. *Exploring metabolic responses to emotional prosody with fNIRS*. Talk given at "Listen and Learn Workshop", Macquarie University, Nov 2019.

CONFERENCE TALKS

1. **Moffat**. *Multi-person and multi-session fNIRS as a tool for studying social connections across generations*. Talk given at Society for Social Neuroscience (S4SN) 2026. Montreal, Canada, Aug 2026.
2. **Moffat**, Dumas, Naudszus & Cross. *Interpersonal dynamics of mitigating loneliness using intergenerational creative art-making*. Talk given at European Society for Cognitive and Affective Neuroscience (ESCAN) 2026, Rome, Italy. Jun 2026.
3. **Moffat** & Cross. *Perspectives on embodiment and enjoyment when watching or engaging in dyadic synchrony*. Talk given at International Conference on Music Perception and Cognition (ICMPC) 2025, São Paulo, Brazil. Jul 2025.
4. Cross & **Moffat**. *Exploring behavioural and neural mechanisms of movement synchrony from performers' and observers' perspectives.* Talk given at Joint Action Meeting X 2025, Turin, Italy. Jul 2025.
5. **Moffat**. *Multi-person fNIRS as a bridge between populations – Insights into intergenerational relationship development*. Talk given at Organisation for Human Brain Mapping (OHBM) 2025, Brisbane, Australia, Jun 2025.
6. **Moffat** & Cross. *Rose-tinted embodiment: Mirroring impacts enjoyment, empathy, and cortical activity when observing synchronous movements*. Talk given at fNIRS UK, Colchester, UK, Sep 2023.
7. **Moffat** & Cross. *Observer's traits predict accuracy of synchrony estimation and enjoyment of dyadic mirror-game movements*. Talk given at the 9th Joint Action Meeting, Budapest, Hungary, Jul 2023.
8. **Moffat**, Caruana & Cross. *Cortical correlates of inhibition when observed by synchronised vs. non-synchronised peers*. Talk given at Neuroscience 2022 (SfN), San Diego, USA, Nov 2022.
9. **Moffat**, Baskent, Luke, McAlpine & van Yper. *Pilot: Processing emotional prosody in normal hearing listeners with fNIRS*. Talk given at Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, USA, Jul 2019.

CONFERENCE POSTERS

1. **Moffat**, Dumas & Cross. *Intergenerational relationships as seen through hyperscanning*. Zurich Social Connection Workshop, Zurich, Switzerland, Sep 2025.
2. Leisten^, **Moffat**, Caruana & Cross. *Towards hands-on social robots for accessible, equitable, and sustainable technology-based learning*. Presented at IEEE RO-MAN in Eindhoven, Netherlands, Aug 2025.
3. **Moffat**, Dumas & Cross. *fNIRS insights into intergenerational relationship development*. Presented at Organisation for Human Brain Mapping (OHBM) 2025, Brisbane, Australia, Jun 2025.
4. **Moffat**, Portier, Häuselmann, Dumas & Cross. *Longitudinal insights into motor coordination and neural synchrony within and between generations.* Presented at fNIRS 2024, Birmingham, UK, Sep 2024.
5. **Moffat**, Portier, Häuselmann, Casale & Cross. *Intergenerational social interactions: How do motor and neural synchrony develop over time?.* Presented at Social & Affective Neuroscience Society (SANS), Toronto, Canada, Apr 2024.
6. Casale^, **Moffat** & Cross. *Arts engagement and embodied experience shape our aesthetic perceptions of socially-intentioned body movements*. Australasian Cognitive Neuroscience Society (ACNS), Sydney, Australia, Nov 2023.
7. Peterson^, Roberts, **Moffat** & Polito. *The relationship between altered states of consciousness, nature relatedness, and pro-environmental behaviour*. 3rd Psychedelic Symposium UW Madison, Madison, USA, Nov 2023.
8. **Moffat** & Cross. *Quantifying observed motor synchrony: Movement predictability and inter-individual traits predict accuracy*. Presented at Experimental Psychology (EPC), Canberra, Australia, Apr 2023.
9. Casale^, **Moffat** & Cross. *Art experience enhances enjoyment of dynamic human movement*. Presented at Australian Society for Experimental Psychology (EPC), Canberra, Australia, Apr 2023.

10. **Moffat**, Caruana & Cross. *Cortical activity evoked by synchronised vs. non-synchronised peer observers as detected with fNIRS*. Presented at fNIRS 2022, Boston, USA, Oct 2022.
11. Caruana, **Moffat**, Blanco & Cross. *A welcome social presence: Attentive and responsive robot reading buddies are preferred*. 8th Australasian Society for Social and Affective Neuroscience (AS4SAN) Conference, Sydney, Australia, Jun 2022.
12. **Moffat**, Baskent, Luke, McAlpine & van Yper. *Cortical responses to vocal emotions with attenuated voice pitch variation measured with fNIRS*. Presented at fNIRS 2021 virtual conference, Online, Oct 2021.
13. **Moffat**, Baskent, Luke, McAlpine & van Yper. *Pilot: Mapping emotional prosody in normal hearing listeners with fNIRS*. Presented at Association for Research in Otolaryngology (ARO), San José, USA, Jan 2020.
14. **Moffat**, Baskent, Luke, McAlpine & van Yper. *Pilot: Using fNIRS to explore emotional prosody perception*. Presented at 12th Speech in Noise Workshop (SPIN), Toulouse, France, Jan 2020.

TEACHING

ETH Zurich

The Cutting Edge of Social Brain Imaging	Fall 2026, Fall 2025, Fall 2024
Mind Meets Machine	Spring 2026, Spring 2025
Learning With and From Robots	Spring 2024, Fall 2023

Macquarie University

Delusions and Disorders of the Mind and Brain (2-3 groups each semester)	Fall 2020, Spring 2020, Fall 2019
Introductory Phonetics and Phonology (2 groups)	Spring 2020
Introduction to Psycholinguistics (1 group)	Fall 2020
Phonological Analysis (2 groups each semester)	Fall 2020, Fall 2019
Language as Evidence (1 group)	Fall 2020

Guest lectures given as part of courses offered at other universities

Human-Machine Interaction, Polytechnic University of Milano	Nov 2026, Nov 2025
Psychophysiological Methods in Psychology and Neuroscience, New Bulgarian University	Nov 2024

Other teaching

Visualising data with ggplot2–Preparing figures for publication and outreach, R Ladies Zurich Workshop Nov 2024

SUPERVISION & MENTORSHIP

PhD Dissertations

ETH Zurich:
Tessa Portier [Nov 2024–present]
Courtney Casale [Sep 2024–present]
Tanja Ulrich [Sep 2024–present]
Fenella Symes [Sep 2023–Dec 2024]

Bachelor/Master's Theses

ETH Zurich (BSc):
Greta Bramow [Jul 2024–May 2025]

ETH Zurich (MSc):
Ahmed Eldably [Sep 2025–present]
Anna Mayr [Jan 2026–Jul 2026]
Filiz Kanele [Oct 2025–Jun 2026]
Leia Jekel [Feb 2026–May 2026]
Luca Naudszus [May 2024–May 2025]
Medea Häuselmann [Feb–Aug 2024]
Tessa Portier [Oct 2023–Apr 2024]

Macquarie University (MRes):
Hannah Cahill [Oct 2022-2024]
Abigayle Peterson [Oct 2022–2023]
Courtney Casale [Sep 2022–2023]

Interns

ETH Zurich:
Chantal Nagel [Jul–Oct 2025]
CJ Abeshaus [Jun–Aug 2025]
Alistair Gadola [Feb–March 2024]
Leonie Roos [Jul–Sep 2023]

Macquarie University:
Jonathon Clare [Apr–Jun 2023]
Annika Richter [Jul–Sep 2022]
Sabrina Diep [Feb–May 2022]

RECENT TRAINING

2026 *MR Safety Training*, ETH Zurich
2026 *Representational Similarity Analysis to Understand How the Brain Represents Information*, University of Zurich
2025 *Hidden labour and activism in open science*. Open Science Retreat, Digital Research Academy
2024 *Integrating sustainability in every field*, ETH Zurich
2024 *Hands-on machine learning with fNIRS using the BenchNIRS Python framework; Cedalion: […] fNIRS & DOT Data Analysis with Machine Learning Approaches,* fNIRS2024 Conference
2024 *Machine Learning for Health Data,* University of Zurich
2022 *Deaf Awareness Training; Mental Health First Aid; Foundations of Supervision*, Macquarie University

RESEARCH POSITIONS

2018 **Laboratory for Oral Language Acquisition (Prof. Aude Noiray) & BabyLAB (Prof. Barbara Höhle)**
*Research Assistant***,** University of Potsdam
Topics: Developmental language disorders; Ultrasound tongue imaging; fNIRS

2016 **Section for Clinical Cognition Sciences (Prof. Ferdinand Binkofski) & Department of Neurology, Stroke/Aphasia Units (Dr. Miriam Abel)**

*Research Assistant***,** RWTH Aachen University Hospital
Topics: TMS; EEG; Language production; Electrical pharyngeal stimulation

2015 **School of Rehabilitation Sciences (Prof. Patricia Roberts)**
Research Assistant, University of Ottawa
Topics: Bilingualism; Verbal fluency; Language production

PROFESSIONAL SERVICE

COMMISSIONS OF TRUST

2026 **Guest Editor:** *Oxford Open Neuroscience* special issue "Toward a Mechanistic Understanding of Interpersonal Synchrony"
2026 **Member of the Board of Directors:** Society for functional Near-Infrared Spectroscopy (SfNIRS)
2024–present **Member of Education Committee:** Society for functional Near-Infrared Spectroscopy (SfNIRS)
2025–present **Co-lead of Working Group:** EU-SABV (Sex as a Biological Variable COST Action)
Focus: Clinician and industry collaboration to enhance the translational value of sex-inclusive research
2025–present **Expert affiliate:** Women's Brain Foundation (WBF) Academy
2024–present **Vice President:** Departmental Scientific Staff Association (ASST)

REFEREEING ACTIVITIES

2023–present **Ad-Hoc Manuscript Review:** *Brain & Behavior, Humanities and Social Sciences Communications; Imaging Neuroscience; Journal of Cognitive Neuroscience; NeuroImage; Neuropsychologia; Oxford Open Neuroscience; PLOS One; PNAS; Quarterly Journal of Experimental Psychology; Review of General Psychology; Social Cognitive & Affective Neuroscience; Translational Psychiatry*

2024, 2026 **External Master's Thesis Examination:** Examination of theses submitted for Master's of Clinical Audiology at Macquarie University; Master in Cognitive Science at Sorbonne Université and Université Paris Cité

2023, 2025 **Federal Grant Review**: Reviewer for *National Science Foundation (NSF) Perception, Action, and Cognition Grant; Austrian Science Fund (FWF) ASTRA Awards*

COMMUNITY OUTREACH

2025 **A Popularity Contest at the Robotic Petting Zoo**, Bits n Bolts Festival, Zurich
Mediated an interactive public art performance, where participants observed robotic vacuums with different personalities and voted for their favorite robots by feeding them colored beads.

2024 **Apero: Brain activity during social interaction**, ETH Zurich
Hosted an event for participants (122) from a longitudinal intergenerational study, where I presented study results and recorded participants' feedback on the study.

2022 **Crestwood Secondary School**, Canada | **Camp Aspire**, Macquarie University
Held workshops for Indigenous Australian [Jan 2022] and Canadian [Oct 2022] high school students about measuring and maximizing the social benefits of motor synchrony.

2021, 2022 **LEAP UP University Experience Day** | **Widening Participation**, Macquarie University
Introduced refugee [May 2021] and Indigenous Australian [Apr 2022] high school students to robotic reading companions and scientific practices used to study child–robot interaction.

MEDIA COVERAGE

2026 Tracking brain waves reveals a surprising twist in how different generations form social bonds. Blog, *PsyPost*
2026 Brain waves and intergenerational social bonds. Podcast, *The Last Show with David Cooper*
2026 Social brain research offers new insights into how strangers become friends. News, *ETH Zurich*
2026 When people of different generations create together, brain activity changes. Press release, *PLOS Biology*
2025 Could reading robots help boost literacy rates in children? Radio interview, *ABC Radio*
2025 These robot cats have glowing eyes and artificial heartbeats – and could help reduce stress in children. Magazine article, *The Guardian*
2025 Can social robots lend a hand with children's reading?. Magazine article, *The Educator*

CONSULTING SERVICES (VOLUNTARY)

2026 **MOSAIC (Multimodal Open-Science AI-Powered Integrative Computation):** Consulted on minimal viable product from researcher perspective.
2025 **DigeHealth:** Consulted on communicating the scientific approach for mobile acoustic monitoring of gut sounds.
2024 **Connectome:** Consulted on feasibility of using fNIRS as the core technology for a healthtech startup.

LANGUAGES

Native – English
Fluent – German, French
Intermediate – Finnish, Croatian, Spanish